1 Detailed planning and assignment functions

Purpose

Use the detailed planning and assignment functions to:

- plan operations for a workplace and to reserve the workplace's capacities
- deallocate operations from a workplace back to a group and as a result, to free up capacities at that workplace
- identify operations as fixed (fix a date and a workplace for the OP) so that they are not replanned by automatic planning functions.

To do this, you can typically plan, replan or deallocate operations using the drag & drop function.

This document outlines the planning functions provided in the Graphic planning.

Integration

If you schedule an operation for a workplace, you define the processing sequence in production. The sequencing list of the shop floor terminal shows this processing sequence.

Requirements

Before you start the detailed planning, you should define and create relevant planning profiles that integrate aspects of organization, planning procedure and competences.

1.1 Manual scheduling using drag & drop

If you use the Shop Floor Scheduling, operations in the ERP system are generally planned for a (capacity) group.

After individual capacities (workplaces) have been opened in the planning board, you can now drag operations from the tabular or graphic pool of groups and plan them onto the workplace if the workplace is not locked. Left-click an operation and drag it to the required location in the graphic planning board.

If you enable the extension graplocking, you can only plan an operation for a workplace if the workplace is not locked.

1.2 Resolving conflicts during planning and replanning

When you plan or replan operations in the Gantt using the drag & drop function, the OP is planned on the date the mouse pointer is on when you "drop" the operation. After dropping the OP, the checks activated in the settings are run for the operation.

- Checking basic dates of operation
  The system checks if the planned start lies ahead of the earliest start of the operation or if the planned end lies past the latest end of the operation.

- Checking preceding/ subsequent relationships
  The system checks if the operation overlaps its preceding or its subsequent operation. Any overlaps are accepted.

  The application can only check preceding and subsequent operations if they are included in the current planning profile.

- Checking the used capacity
  The system checks if planning results in any capacity overloads. The system checks the workplace where the operation has been/ is being planned and the production resources and tools assigned to the operation, provided that they have been defined as resources in the system (depending on license).

- Checking the capacity of required staff (depending on license)

- Checking the required staff's qualifications (depending on license)

- Checking material availability (depending on license)

The checks activated in the settings are processed in sequence. If a user action (planning/ replanning) is not allowed following one of the above-mentioned consistency checks, then the action is canceled. The remaining checks are not performed. The conflict is displayed in a pop-up window. The planner can decide which steps can be taken to resolve the conflict:

If a validation check is deactivated, resulting conflicts are not displayed, i.e. a possible conflict is implicitly confirmed. But the conflict list still shows the conflict.

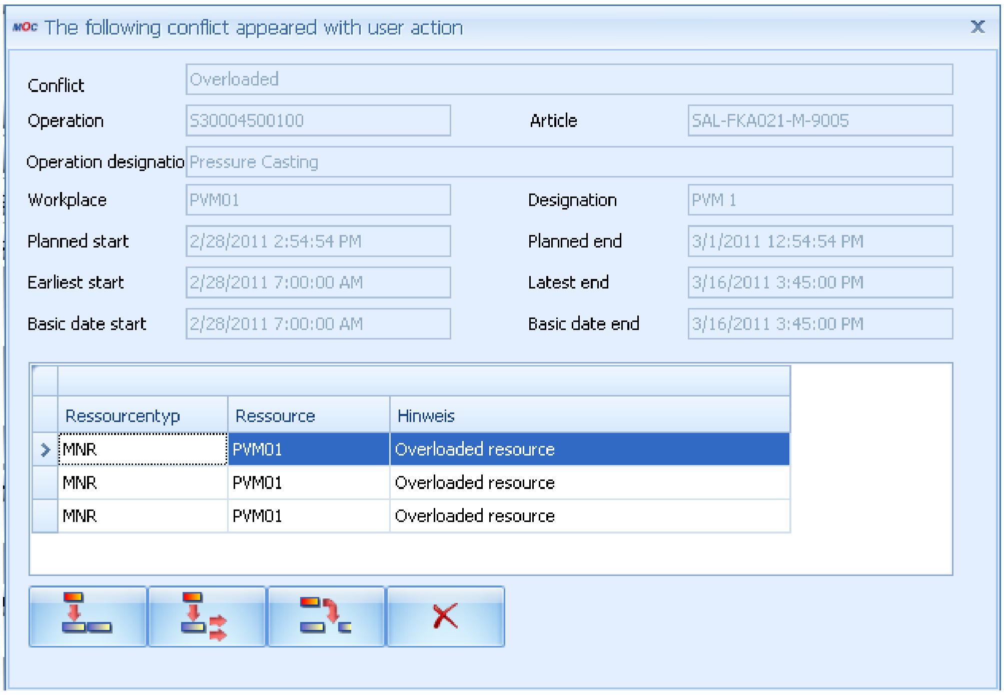
Plan / move beyond time frame

The conflict list indicates:

- If planning an operation for a workplace or moving it causes the planned start of the operation to be ahead of the earliest start of the operation or ahead of the order's basic start date or

- if planning an operation for a workplace or moving it causes the planned end of the operation to lie beyond the latest end of the operation or beyond the order's basic end date

when the operation is "dropped" (for further information on the fields: see below):



→→

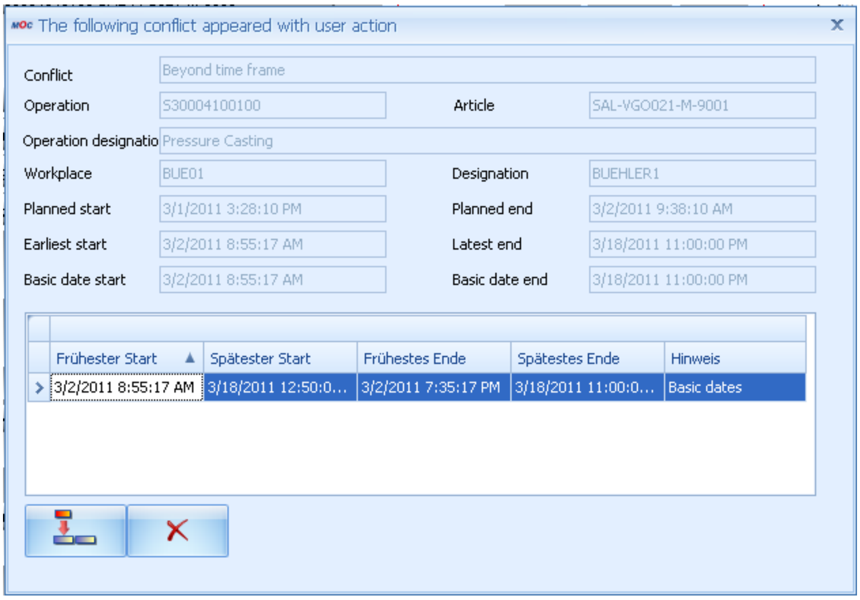
Operation-related dates, e.g. earliest start time (EST), latest start time (LST), earliest end time (EET) and latest end time (LET) must either be transferred from the ERP system or, alternatively, can be determined during the lead time scheduling process.

Possible reactions

	Plan anyway	The operation is planned nonetheless.
	Cancel	The planning process is canceled.

Violation of relationships

What might also be the case is that the operation has a preceding and/or a subsequent operation with planning dates that are not consistent with those of the currently planned operation. A dialog opens to notify you if the required date overlaps those dates (for further information on the fields: see below:



The list shows the operation coinciding with the currently planned operation. The list also shows the planned start, planned end, and the workplace of the already planned operation. The text displayed in the column *Error* describes the conflict as seen from the already planned operation.

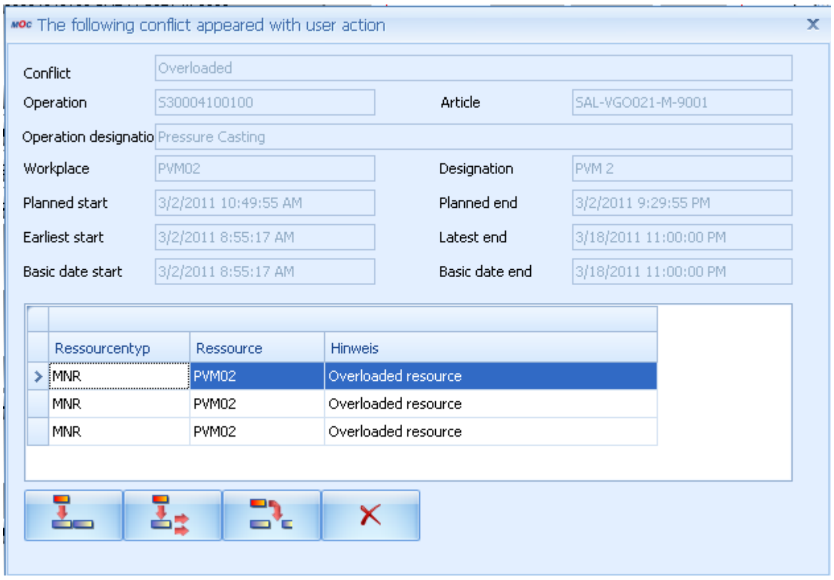
Possible reactions

	Plan anyway	The operation is planned nonetheless.
	Move predecessor/ successor	An attempt is made to move the already planned predecessor or successor so that there is no more conflict between the two adjacent operations.
	Move the subsequent planning scenario.	**Please note:** This function only integrates the machine assignment. You can use this function for a maximum of 150 operations that are planned for a workplace.
	Cancel	The planning process is canceled.

Checking if resources are overloaded (overallocated)

When an operation is planned, the system checks if this will overload resources. On the one hand, the workplace where the operation is planned is a resource. On the other, the system also checks secondary resources that are needed for production. Secondary resources are defined for the operation in the list of *production resources and tools* and identified in the WRM module as relevant to assignment.

This dialog indicates if a conflict occurs (for further information on the fields: see below):



The list shows both operations (the planned one and the one to be planned). The list also shows the planned start, planned end and the resource that is assigned twice.

Possible reactions:

	Plan anyway	The operation is planned for this date nonetheless; as a result, the resource is assigned multiple times.
	Search for free gap	The operation is planned in the first suitable gap.
	Move the subsequent planning scenario.	**Please note:** This function only integrates the machine assignment.
	Cancel	The planning process is canceled.

Checking staff availability

 This check is only available if you enable the extension graptcpq.

The qualifications required by the operation are compared with the assigned staff's qualifications. A conflict message "Staff shortage (qualification)" pops up if a required qualification cannot be met.

If qualifications are basically met, i.e. employees with the required qualification are available at the workplace where the operation is planned, the system now verifies the number of required qualifications. If the required number is not available, the conflict message "Staff shortage (capacity)" appears.

Possible reactions

	Plan anyway	The operation is planned nonetheless.
	Cancel	The planning process is canceled.

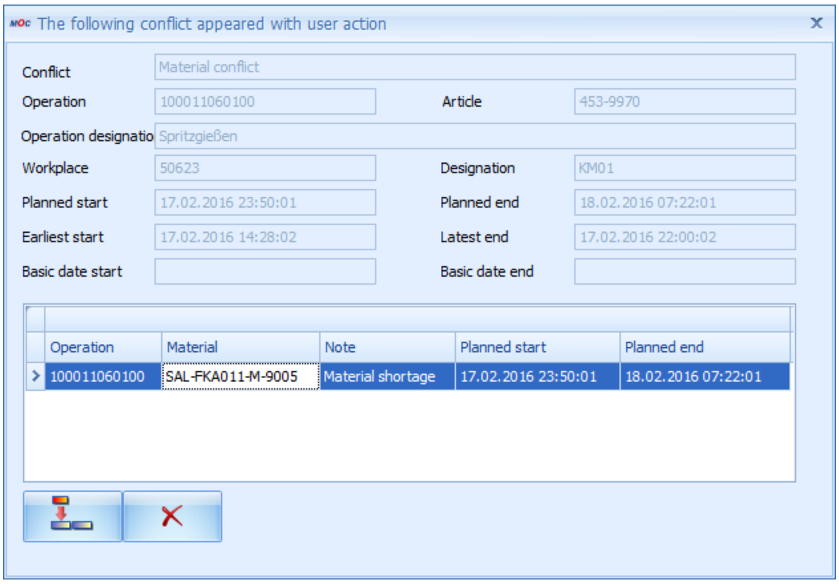
You can find further information about how to check if personnel is available here.

Check material availability

This check is only available if you enable the extension grapvemvp and set the option "check material availability" in the settings.

The material availability check informs you if sufficient material is available to produce the operation. This check makes sure that sufficient material is always available for production.

When you plan an operation for a workplace, a conflict dialog appears if the system identifies that the material required for the operation is not available in sufficient quantities:



Possible reactions

	Plan anyway	The operation is planned nonetheless.
	Cancel	The planning process is canceled.

Material availability is only checked if the detail view planned inventory levels is shown. The system only checks material with a valid ATP inspection group assignment.

The system only checks if material is available if planning is performed manually.

Information about dialog fields:

Operation: MES order number of the operation

Article: is taken over from the operation.

OP name: is taken over from the operation.

Workplace, name: workplace where an attempt is made to plan the operation.

Planned start: date when an attempt is made to plan the operation.

Planned end: calculated by the Shop Floor Scheduling and based on planned start + additional setup time + setup time + remaining run time + retooling time.

Earliest start time (EST), latest end time (LET): these fields relating to operations must either be transferred from the ERP system or, alternatively, can be determined during the lead time scheduling process.

Basic start date/basic end date: basic dates relating to the order (header) transferred from the ERP system.

1.3 Fixing operations

Fix operation

Use this function to fix the selected operation(s) and identify these OPs as fixed. The settings specify how fixed operations are visualized.

Already fixed operations remain fixed. If you attempt to fix already fixed operations, the system does not interpret this as a changed planning.

You cannot fix operations that are included in the pool of groups.

You can neither fix nor unfix logged on operations.

You cannot deallocate or replan fixed operations to a different workplace.

Unfix operation

Use this function to unfix the selected operation(s) and to remove the visual identification.

Operations that are not fixed remain so. If you attempt to unfix operations that are not fixed, the system does not interpret this as a changed planning.

You cannot unfix logged on operations (since you cannot fix them either).

If you enable the extension graplocking and the operation is planned on a locked workplace, you cannot fix/unfix the operation.

1.4 Close gaps

You can use this function to close any gaps between individual operations of a workplace.

All of the planned operations to the right of the planning lead time are shifted to the left to join up with the immediately preceding operation at the workplace, while maintaining the chronological order of operations defined for the workplace. The workplace must be available. You can only plan operations where the workplaces are available as per the shift calendar.

The application derives the sequence for operations located on the "now" line from the planned (start) dates stored in the database. The application integrates these dates when closing gaps so that the originally planned sequence is maintained. Exception: If the planned dates of these delayed operations are identical in the database (e.g. if the OPs are on the "now" line and you save the planning), then the application can no longer identify the original sequence. In this case, the sequence, i.e. which OP is considered first, is "random" (not prioritized).

Operations are moved up to the planning lead time at most.

Fixed operations are not moved.

The system integrates relationships with preceding operations.

The planning can still include gaps that might be closed even though you used the *close gaps* function. This is the case if more than nine operations are planned and connected via an order network.

This can still result in gaps.

If you enable the extension graplocking, the function is not supported if you want to execute the function for a locked workplace.

1.5 Replan an operation for another group

HYDRA differentiates between

- **Control**

  understood as including the actual detailed planning tasks in Shop Floor Scheduling, such as allocation to workplaces/ machines, resource checking, sequencing etc.

  and

- **Planning**

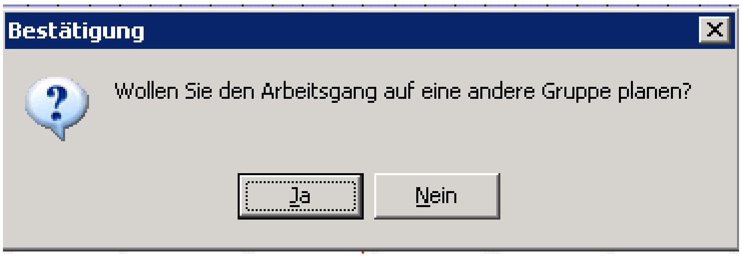
  understood as intervening in the structure of the order/ work plan. These tasks are generally understood as ERP functions.

You can replan an operation onto another capacity group or onto a workplace of another group in the graphic planning board, of course, but this is not understood to be a planning function in the sense of the above definition. Therefore, please note the following:

- Process times are not recalculated during replanning.

- Assigned resources (components, production resources and tools) remain assigned.

- The application does not check if replanning makes sense (e.g. you replan a drilling operation for a "milling" group).



If you enable the extension graplocking, the function is not supported if you want to execute the function for a locked workplace or a locked group.

1.6 Deallocate operation

If you deallocate (unplan) an operation, the application returns the operation to the pool of groups. Depending on the option "Show OPs in the pool of groups of the Gantt", you need to take the following steps:

The option "Show OPs in the pool of groups of the Gantt" is set

To **deallocate** (unplan) an operation, select the required operation in the planning board and then drag and drop it, i.e. "move" it back into the graphic pool of groups in the planning board.

The operation is displayed at the scheduled start time in the pool of groups of the planning board (not necessarily the same place where you "dropped" the operation).

The option "Show OPs in the pool of groups of the Gantt" is not set

To **deallocate** (unplan) an operation, select the required operation in the planning board and move (drag and drop) it into the graphic pool of groups in the planning board (not into the tabular pool of groups).

When you save planning, the system sets the option "planned" to "group" for deallocated operations. The workplace where the operation had been previously planned is NOT deleted.

If you enable the extension graplocking, you can only deallocate (unplan) an operation if the following conditions are met:
- the workplace where the OP is removed must not be locked
- the group where the OP is added must not be locked.

1.7 Automatic assignment in graphic planning

Function authorization	grapt.autopl

If you click the button , you can cause the system to make an automatic assignment. If you use the function, you can differentiate between whether:

- only unplanned operations should be planned.
  If you use the option "Plan OPs from group level only", you plan operations that are currently not planned.

- only selected operations should be planned.

  This option is only available if you enable the extension graptsbap.

  If you enable the option "plan selected operations only", the planning function only plans those operations you have selected in the pool of groups. This means that any operations already planned remain so.

  You can select operations either in the tabular pool of groups or the graphic pool of groups. You can use the Ctrl key to select single operations.

  The automatic planning function does not plan operations that cannot be planned due to existing conflicts.

- everything should be replanned.
  If you select the option "Entirely replan all OPs", the application deallocates (unplans) all operations that are not fixed, and then runs the planning algorithm.

The entire planning is considered with either action.

The planning algorithm utilizes the planning strategy (priority rule) defined in the currently selected planning variant and the selected capacities.

After automatic assignment has been completed, the system shows the results of this automatic assignment. The system shows how many OPs have been planned. A list includes the operations that had conflicts, and could therefore not be planned.

If the option *Fix operations in planning time fence after automatic assignment* is set in the settings, the system automatically fixes operations the planned start date of which is within the planning time fence after automatic assignment.

If you enable the extension graplocking, you can only execute the function if all workplaces and groups pertaining to the planning profile are not locked.

1.8 One-step planning

Function authorization	grapt.sbsp

Use the button "plan operation" to carry out "one-step planning".

Once you've clicked this button, the next matching operation included in the pool of groups is selected and planned. This process depends on the rules specified in the planning variant you selected while requesting data. You can reproduce automatic planning step by step if you click the button multiple times.

The operation that is planned at last is selected in the Gantt chart. Consequently, you can simply deallocate (context menu) or replan the OP to another position.

If you clicked the button, but no operation could be planned, you are informed about it. A pop-up window appears indicating *"No further operations can be planned"*.

You have to enable the extension grapt.sbsp in order to use the function.

If you enable the extension graplocking, you can only execute the function if all workplaces and groups pertaining to the planning profile are not locked.

1.9 Individual shift times

Define individual shift times to specify for a workplace and a certain period (date/time from/to) whether the time considered is working time or non-working time.

The configuration is described here.

Create individual shift time via context menu of workplace

You can right click a workplace or a group in the graphic planning (Gantt chart) to open the dialog for creating individual shift times. Once you have entered the data, the additional shift times are integrated directly in the planning board.

The suggested period is preset as:

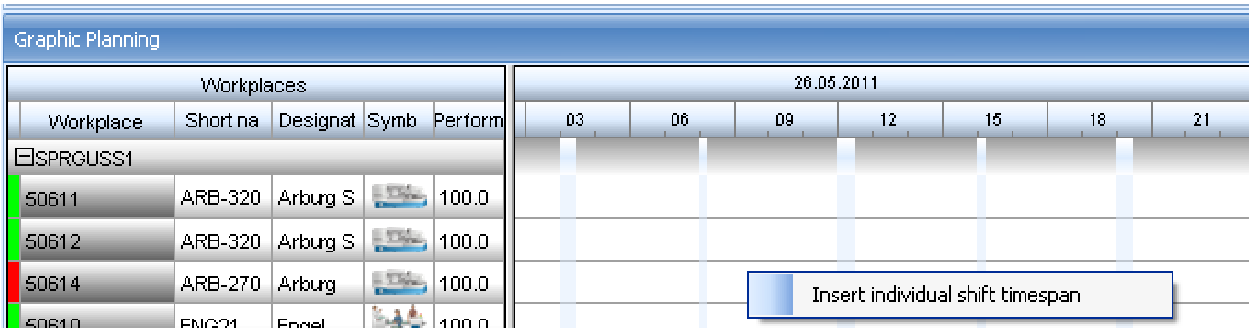
From: "now" (the point in time when you called this function)

To: the point in time currently visible on the far right in the planning board.

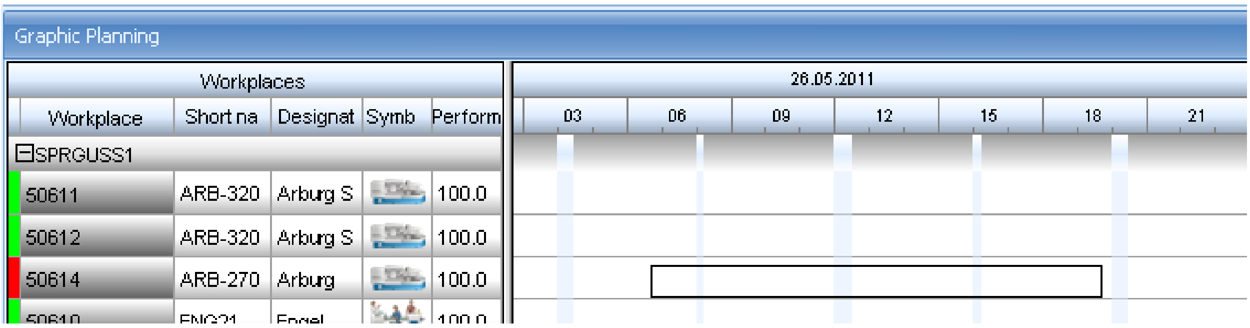
Depending on the type of period, i.e. working time or not, the color of the period is either white or yellow. You can still change colors.

Using the mouse to select a period

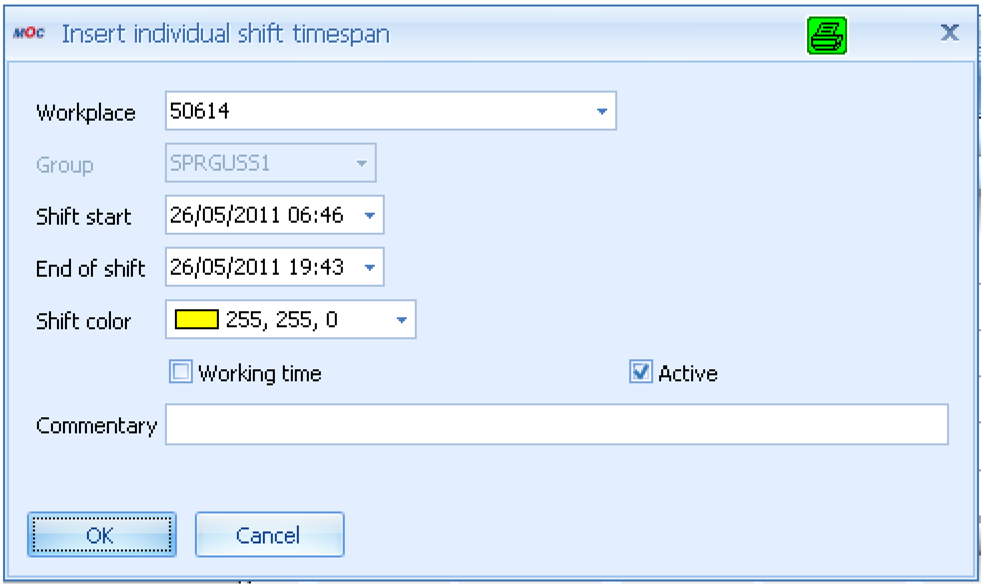
Position the mouse in the planning board chart and right click to activate the "selection mode".



Now position the mouse over the workplace for which you would like to define the individual shift time; continue to keep the mouse button pressed and move the mouse to the right until you reach the required time domain, then release the mouse button.



A window will then open where you can create individual shift times. The selected time domain is preset by default.



Depending on the type of period, i.e. working time or not, the color of the period is either white or yellow. You can still change colors. You must check the *"active"* checkbox so that the entry will be used in planning.

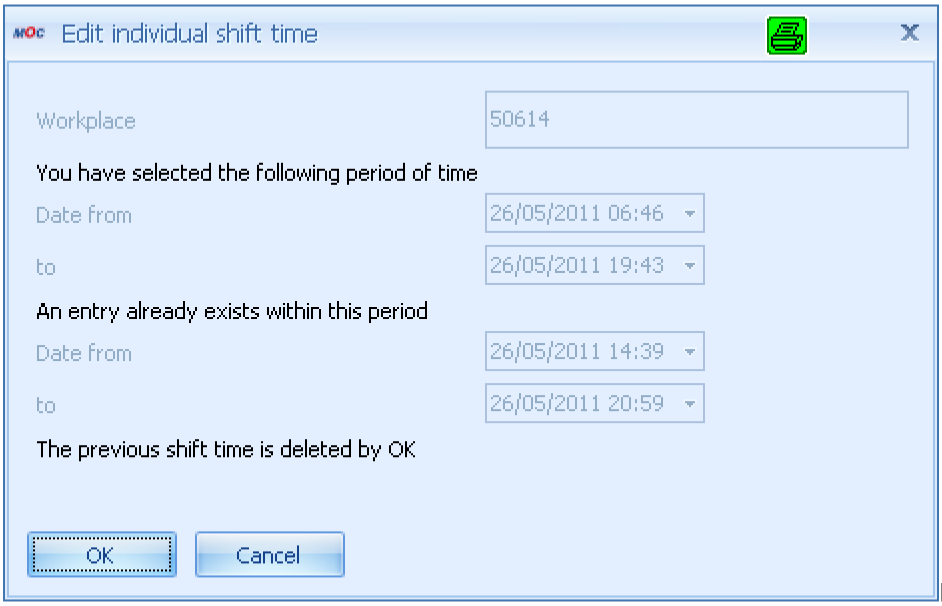
The workplace assigned by default is always the last workplace entered, even if the selected time domain was dragged across several workplaces. However, you can still select more workplaces from the list manually.

The selection mode is reset after each action.

You should not use the color *turquoise* if you enable the extension graptcpq. As this extension uses *turquoise* to show personnel availability. You can find further information about how to check if personnel is available here.

Once you've clicked OK, data is verified and transferred to the planning board.

You can add only one entry for each period of a workplace. When you attempt to add an entry and the system detects that an entry already exists for this period, you can either delete the previous entry or cancel your current entry:



If multiple individual shift times exist, this dialog opens for each existing shift time, and you can decide whether or not to delete the entries.

Individual shift times are not committed to the database until planning has been saved.

These additional shift times are only used in Shop Floor Scheduling, not as part of data collection.

1.10 Create and display notes about the operation

You can view, enter and delete notes for the selected operation.

Function authorization	edopnote*

General display

A detail application shows the operation notes. Click the corresponding toolbar icon to enable the detail application.

The table includes the following columns: Short text, modified by, date, time, display at terminal, MES order number. The memo text is displayed to the right of that. Between table and memo is a splitter that is saved.

If the detail application is visible and you click one or more OP bars, the detail application contents are updated.

Add new notes

Proceed as follows to add a new note:

- use the context menu of the operation bar
- use the corresponding button in the "Operation notes" index tab of the toolbar; the index tab is active/ visible if you click the detail application "Notes".

You must specify a short text and a long text in the input dialog.

The detail application "Notes" shows the note, once you have saved the note by clicking ✓. The operation bar now shows a respective icon indicating that a note is available.

Edit a note

You can edit an existing note. To do this, take the following steps:

- If the index tab "Operation notes" is not visible in the toolbar, click on the detail application "Notes".
- Select the note you would like to edit in the detail application.
- Click the button "Edit note" to open a dialog displaying existing information.
- After modifying the data, click ✓ to confirm your entry.

Delete a note

Edit a note

You can delete an existing note. To do this, take the following steps:

- Click on the detail application "Notes" if the index tab "Operation notes" is not visible in the toolbar.
- Select the note you would like to delete in the detail application.
- Click the button "Delete note". Confirm the confirmation prompt and the note will be deleted.

If no other notes are assigned to the operation, the icon disappears.

Visualization with the operation bar

The icon is shown at the front of the bar. The icon is displayed for operations that are included in the graphic pool of groups or in the Gantt chart .

1.11 Generate campaign

Function authorization	grap.cpnbuild

This document describes campaign production.

1.12 Overlapping

You commonly define dependencies between operations (OP) as finish-to-start relationships. When planning operations, use the finish-to-start relationship to make sure that the successor OP cannot be started (change from status *prepared* to status *running*) until the predecessor OP is finished 100 % (status *finished*).

You can change existing dependencies using the "overlapping" function. Use the *overlapping* function to plan the successor OP even if the predecessor OP has not yet been finished 100 % and is still in status *running*.

This document illustrates how to configure the overlapping function between operations.

1.13 Locking mechanism if used by multiple users

If multiple users use the graphic planning application simultaneously, this function makes sure you cannot change planning if another user is currently making changes to the planning.

The function described in this section is only available if you enable the extension graplocking.

Concept

The locking mechanism checks if objects are locked:
- at cyclic intervals and
- every time before a planning activity is performed

Upon requesting data, the system checks which objects, i.e. workplaces or groups are currently locked by other users. These objects are disabled and identified as locked in the planning board. Disabling such objects ensures that planning cannot be changed for these disabled objects.

But you can still change planning for all objects that are not locked/disabled. At cyclic intervals (by default 120 seconds) the system sends a new synchronization request in the background asking for current locks and disabling further objects, if necessary.

If another user unlocks an object, the current user cannot automatically access this object. The user first has to request data.

If you plan something, the system verifies if there are locked objects. All involved objects are checked. If you want to plan an operation from the pool of groups on a workplace, the system checks both, i.e. if the pool of groups and the workplace are locked. If the required objects are not locked, they will be locked and planning is continued.

If, however, the objects are already locked, planning is rejected and the objects involved are disabled. This prevents you from making changes to planning. You can only change objects and planning, once you've requested data.

But requesting data does not necessarily unlock objects, as objects remain locked until the other user unlocks the objects, e.g. by saving the planning.

Only the user locking an object can actually use this object.

Objects able to be locked

These objects can be locked in graphic planning:

- Individual workplaces or their pool of workplaces
  Automatically locked, once an attempt is made to plan the workplace.
- Groups or pool of groups
  Automatically locked, once planning affects the pool of groups. Example: you want to plan an operation from the pool of groups or you want to deallocate an operation and return it to the pool of groups.

A pale red ( ) background color indicates objects locked by another user. A light green ( ) background color indicates objects locked by yourself. The color is shown for the following columns in the list of workplaces (to the left of the Gantt chart):

- Short name
- Designation (name)
- Group
- Performance level

Request data

If you made changes and locked workplaces or groups, the following confirmation prompt appears, once you request data: "Do you want to release the locked workplaces and workplace groups?"

Answer the prompt by choosing "yes" or "no":

- "Yes": all locks are removed.
- "No": the locks remain active; you can still change your locked objects.

Existing locks are removed automatically after changing the planning profile and refreshing data.

Save planning

If you made changes and locked workplaces or groups, the following confirmation prompt appears, once you save planning (by clicking the option "save planning"): "Do you want to release locked workplaces and workplace groups?"

Answer the prompt by choosing "yes" or "no":

- "Yes": all locks are removed.
- "No": the locks remain active; you can still change your locked objects.

Load planning

You can only load a stored planning if the included workplaces and workplace groups are not locked or if you locked these objects.

"Lock all" button

Click the button "lock all" in the "general" tab of the toolbar to lock all displayed workplaces and workplace groups at once.

The button is no longer shown if the locking mechanism is disabled.

This button does not require a special function authorization.

Automatic assignment

When you call up the function "automatic assignment", the system checks if all objects (groups, workplaces) existing in the current planning profile are available and locks them. In case at least one object is already locked by another user, the system rejects automatic assignment and issues the following message: "automatic assignment can only be used if all displayed workplaces and workplace groups are not locked by another user."

One-step planning ("plan operation")

When you call up the function "plan operation" (button), the system checks if all objects existing in the current planning profile are available and locks them. You cannot use the function if at least one object is locked by another user.

Simulation mode

The locking mechanism is not enabled, if you use simulation mode. In simulation mode the system neither checks if objects are locked nor locks objects.

Existing locks are removed automatically when you switch to simulation mode.

Exit application

Existing locks are automatically removed, when you exit graphic planning.

Unlock manually

As a general rule, all of your locks are removed when you exit the client. You can use the application "locked data records" to unlock specific objects or to unlock all objects. This function is useful, in case you could not save your changes e.g. due to system failure.

Disable the locking mechanism

If the authorization key graplocking is enabled, you can use the following INI configuration to **disable** the locking mechanism:
Name: HLS
MOC user: 0
Section: GRAP
Key: LOCKING
Value: FALSE
Active: ☑

The button is no longer shown if the locking mechanism is disabled.

Limits of locking mechanism

Even though, the locking mechanism has been implemented, it might be the case that the "last user's changes take priority", for example, if the synchronization interval is too large and short-term planning activities are required. The following example illustrates this issue:

There are two workplaces 1 and 2 and two planners A and B. Both planners initially loaded the same data. Operation 1 and 2 are planned on workplace 1; operation 3 is planned on workplace 2.

1. Both planners load data at the time t1.
2. Planner A changes the order of operation 1 and operation 2 on workplace 1. Consequently, workplace 1 is locked.

3. Planner B wants to replan operation 2 from workplace 1 to workplace 2. But replanning is rejected as planner A still locks workplace 1. Workplace 1 is locked for planner B.
4. Planner A saves the planning; the lock is removed.
5. Planner B requests data and is informed that workplace 1 is no longer locked.
6. Planner B replans operation 1 from workplace 1 to workplace 2 and therefore locks workplace 1 and 2.
7. Planner B saves the planning; the locks are removed.
8. Planner A did not notice that. The next synchronization run will only take place at a later time t2. In the meantime planner A had not made changes to the plan and, therefore, was not informed that workplace 1 and 2 were locked.
9. Planner A saves the plan once more and overwrites the changes made by planner B.

Please bear in mind that other user's plans can be overwritten if you replan and save your planning between two synchronization runs.

If you reduce the interval between two synchronization runs, you can minimize the risk of overwriting data.

The default interval for checking if objects are locked is 120 seconds. Use the following INI configuration to change this interval:

Name: HLS

MOC user: 0

Section: GRAP

Key: LOCKING_INTERVAL

Value (e.g.): 60